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1 Proof of the Comparison Test

Since part (b) of the Comparison Test is the contrapositive of part (a), they are logically equivalent and hence it is sufficient to prove part (a). The proof depends on the following lemma.

Lemma 2.1. Let $G(t)$ be an increasing function on an interval (a,∞). Assume there exists $M > 0$ such that $G(t) \leq M$ for all $t \in (a,\infty)$. Then $L := \lim_{t\to\infty} G(t)$ exists and $L \leq$M.

1. We now prove Lemma 2.1.

   Proof. (a) Let $S := \{G(t) : t > a\}$. By assumption, M is an upper bound for S. By completeness of $\mathbb{R}$, $\gamma = \sup S$ exists. Let $\varepsilon > 0$. Show that there exists $t_0 > a$ such that $\gamma - \varepsilon < G(t_0)$.

   (b) Now show that

   $$\gamma - \varepsilon < G(t_0) < G(t) \leq \gamma < \gamma + \varepsilon \text{ for } t > t_0.$$

   (c) Show that (b) implies $|G(t) - \gamma| < \varepsilon$ for all $t > t_0$, which is to say that $\lim_{t\to\infty} G(t) = \gamma =: L$.

   (d) Finally, show that $L \leq M$.

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2. We now prove part (a) of the Comparison Test.

Proof. Consider the functions $F(t) = \int_a^t f(x)dx$ and $G(t) = \int_a^t g(x)dx$, defined for $t \geq a$.

(a) Show that F and G are both increasing on $[a, \infty)$.

(b) Show that $G(t) \leq F(t)$ for all $t > a$.

(c) By assumption $\int_a^\infty f(x)dx$ converges to some number M. By definition of the improper integral, $M = \lim_{t\to\infty} F(t)$. Show that $F(t) \leq M$ and $G(t) \leq M$ for all $t \geq a$.

(d) Apply Lemma 2.1 to complete the proof.

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2 Proof of the Limit Comparison Test

We now prove the Limit Comparison Test.

Proof. 1. Show that there exists $x_0 \in [a, \infty)$ such that $|\frac{f(x)}{g(x)} - c| < \frac{c}{2}$ for all $x > x_0$.

2. Show that the above inequality implies that, for $x > x_0$,

$$0 < \frac{c}{2}g(x) < f(x) < \frac{3c}{2}g(x). \tag{1}$$

3. Apply the Comparison Test to (1) to show that $\int_{x_0}^{\infty} f(x)dx$ converges if and only if $\int_{x_0}^{\infty} g(x)dx$ converges.

4. Conclude that $\int_a^{\infty} f(x)dx$ converges if and only if $\int_a^{\infty} g(x)dx$.

□